

21. Write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Aim

To write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame with a character column.
3. Display the original DataFrame.
4. Use the `str.swapcase()` method on the specified column.
5. Display the modified DataFrame with swapped cases.

Program

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['John', 'ALICE', 'mArK'],
    'City': ['New York', 'LONDON', 'pArIs']
})

print("Original DataFrame:")

print(df)

df['Name'] = df['Name'].str.swapcase()

print("\nDataFrame after swapping case of 'Name' column:")

print(df)
```

Output

```
===== RESTART: C:/Users/shaik/Downloads/query processing/q21.py =====
Original DataFrame:
   Name      City
0  John  New York
1 ALICE   LONDON
2  mArK    pArIs

DataFrame after swapping case of 'Name' column:
   Name      City
0  jOHn  New York
1  alIce   LONDON
2  MaRk    pArIs
>> |
```

Result

Thus, the Pandas program to swap the cases of a specified character column in a DataFrame was successfully executed and the output was verified.

Experiment No: 22

Aim

To write a Python program to draw a line with suitable labels on the X-axis, Y-axis, and a title using Matplotlib.

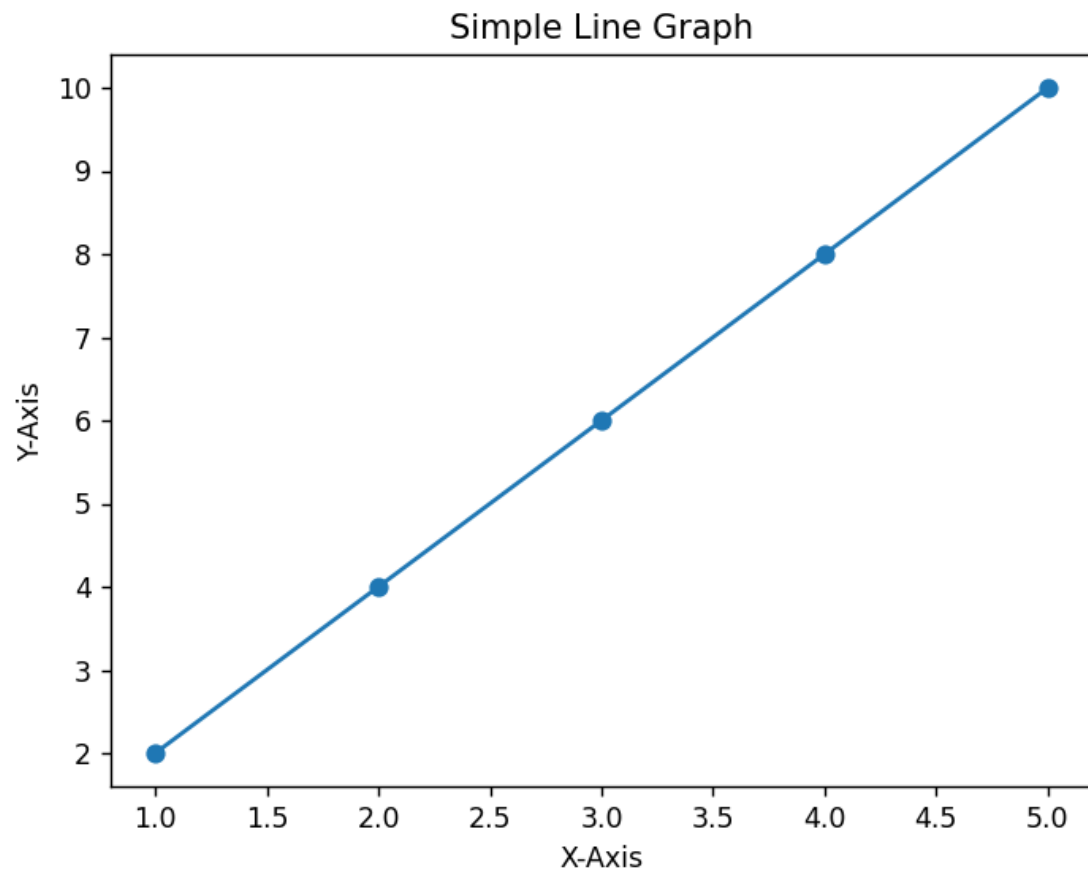
Algorithm

1. Import the matplotlib.pyplot module.
2. Define the X-axis and Y-axis values.
3. Use plt.plot() to draw the line.
4. Add labels to the X-axis and Y-axis using plt.xlabel() and plt.ylabel().
5. Add a title using plt.title().
6. Display the graph using plt.show().

Program

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]
plt.plot(x, y, marker='o')
plt.xlabel("X-Axis")
plt.ylabel("Y-Axis")
plt.title("Simple Line Graph")
plt.show()
```

Output



Result

Thus, the Python program to draw a line with suitable labels on the X-axis, Y-axis, and a title was successfully executed and the graph was displayed.

EXP 23:

Aim

To write a Python program to draw a line using given axis values taken from a text file, with suitable labels on the X-axis, Y-axis, and a title.

Algorithm

1. Create a text file containing X and Y values.
2. Import the matplotlib.pyplot module.
3. Read the data from the text file.
4. Store the values in separate X and Y lists.
5. Plot the data using plt.plot().
6. Add labels to the X-axis and Y-axis.
7. Add a title to the graph.
8. Display the graph using plt.show().

Program

test.txt

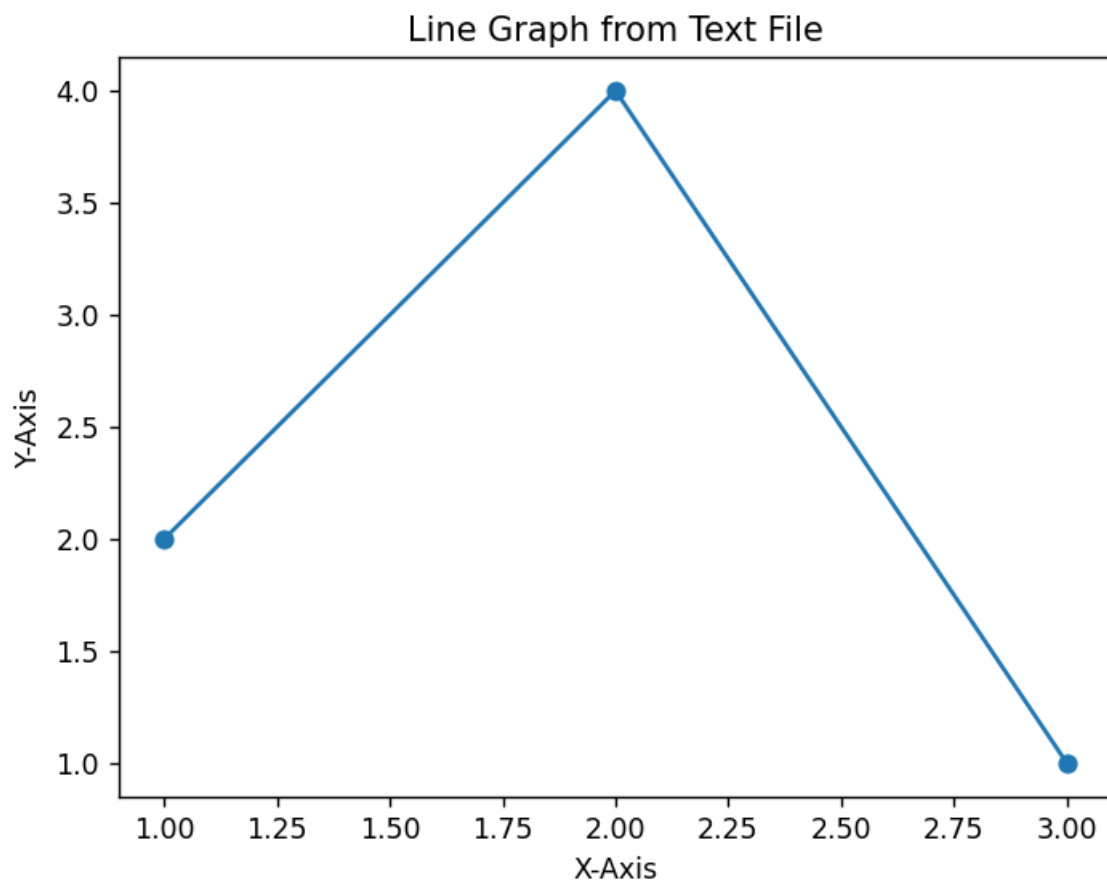
```
1 2  
2 4  
3 1
```

Python Code

```
import matplotlib.pyplot as plt  
x = []  
y = []  
with open('test.txt', 'r') as file:  
    for line in file:  
        a, b = map(int, line.split())  
        x.append(a)
```

```
y.append(b)
plt.plot(x, y, marker='o')
plt.xlabel("X-Axis")
plt.ylabel("Y-Axis")
plt.title("Line Graph from Text File")
plt.show()
```

Output



Result

Thus, the Python program to draw a line using axis values taken from a text file with suitable labels and title was successfully executed and the graph was displayed.

EXP 24:

Aim

To write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 and October 7, 2016.

Algorithm

1. Create a CSV file containing the financial data.
2. Import the pandas and matplotlib.pyplot modules.
3. Read the CSV file using pandas.
4. Extract the Date and Close columns.
5. Plot the data using plt.plot().
6. Add labels to the X-axis and Y-axis.
7. Add a title to the graph.
8. Display the graph using plt.show().

Program

fdata.csv

Date,Open,High,Low,Close

10-03-16,774.25,776.065002,769.5,772.559998

10-04-16,776.030029,778.710022,772.890015,776.429993

10-05-16,779.309998,782.070007,775.650024,776.469971

10-06-16,779,780.47998,775.539978,776.859985

10-07-16,779.659973,779.659973,770.75,775.080017

Python Code

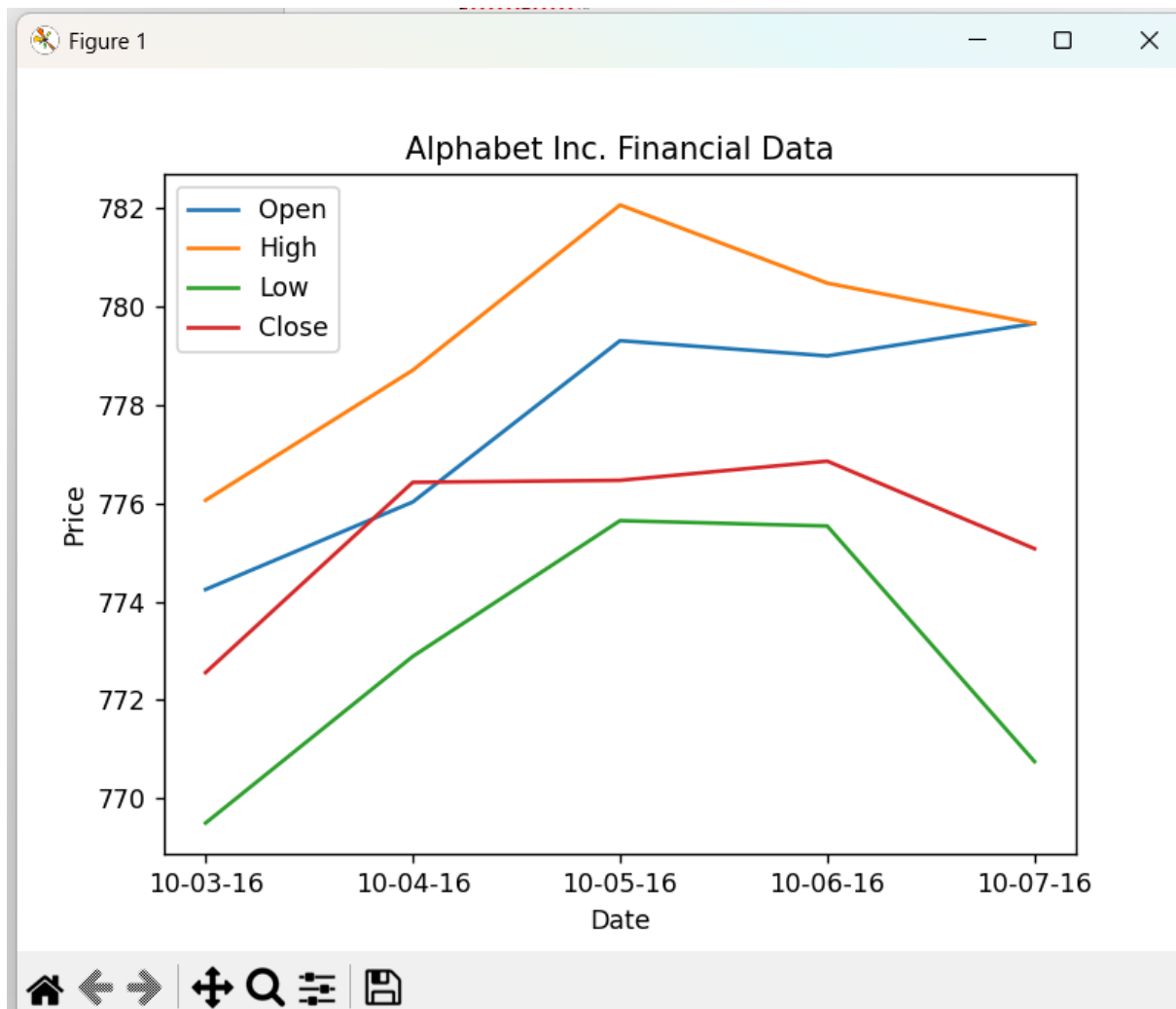
```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
data = pd.read_csv("fdata.csv")
```

```
plt.plot(data["Date"], data["Open"], label="Open")
plt.plot(data["Date"], data["High"], label="High")
plt.plot(data["Date"], data["Low"], label="Low")
plt.plot(data["Date"], data["Close"], label="Close")
plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Alphabet Inc. Financial Data")
plt.legend()
plt.show()
plt.show()
```

Output



Result

Thus, the Python program to draw a line chart of the financial data of Alphabet Inc. between October 3, 2016 and October 7, 2016 was successfully executed and the graph was displayed.

EXP 25

Aim

To write a Python program to plot two or more lines with legends, different widths, and colors.

Algorithm

1. Import the matplotlib.pyplot module.
2. Define the X-axis values.
3. Define the Y-axis values for two lines.
4. Plot the first line with a specific color and width.
5. Plot the second line with a different color and width.
6. Add labels for the X-axis and Y-axis.
7. Add a title to the graph.
8. Display the legend.
9. Show the graph using plt.show().

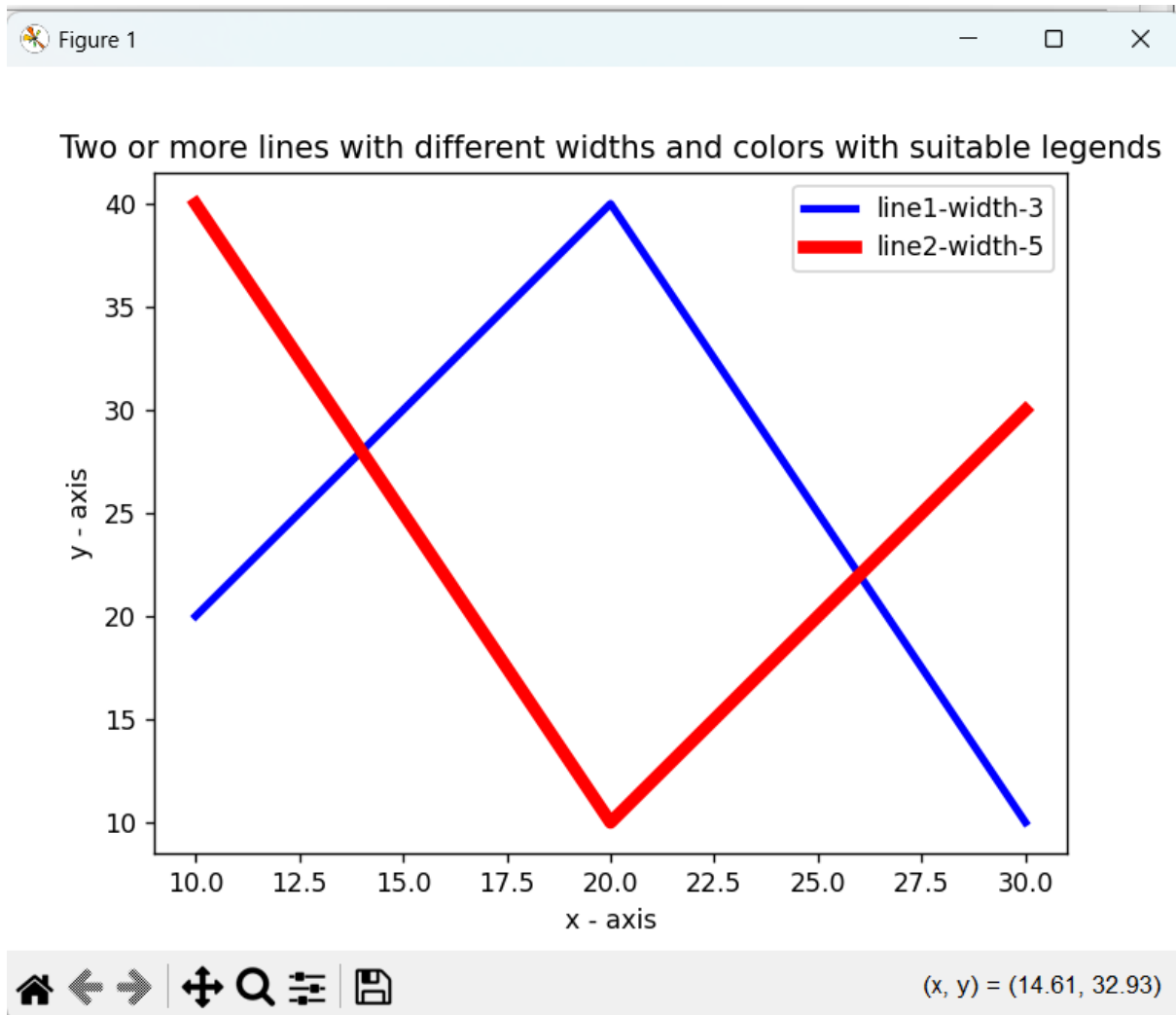
Program

Python Code

```
import matplotlib.pyplot as plt
x = [10, 20, 30]
y1 = [20, 40, 10]
y2 = [40, 10, 30]
plt.plot(x, y1, color='blue', linewidth=3, label='line1-width-3')
plt.plot(x, y2, color='red', linewidth=5, label='line2-width-5')
plt.xlabel('x - axis')
```

```
plt.ylabel('y - axis')
plt.title('Two or more lines with different widths and colors with suitable legends')
plt.legend()
plt.show()
```

Output



Result

Thus, the Python program to plot two or more lines with legends, different widths, and colors was successfully executed and the graph was displayed.

EXP 26

Aim

To write a Python program to create multiple plots in a single figure.

Algorithm

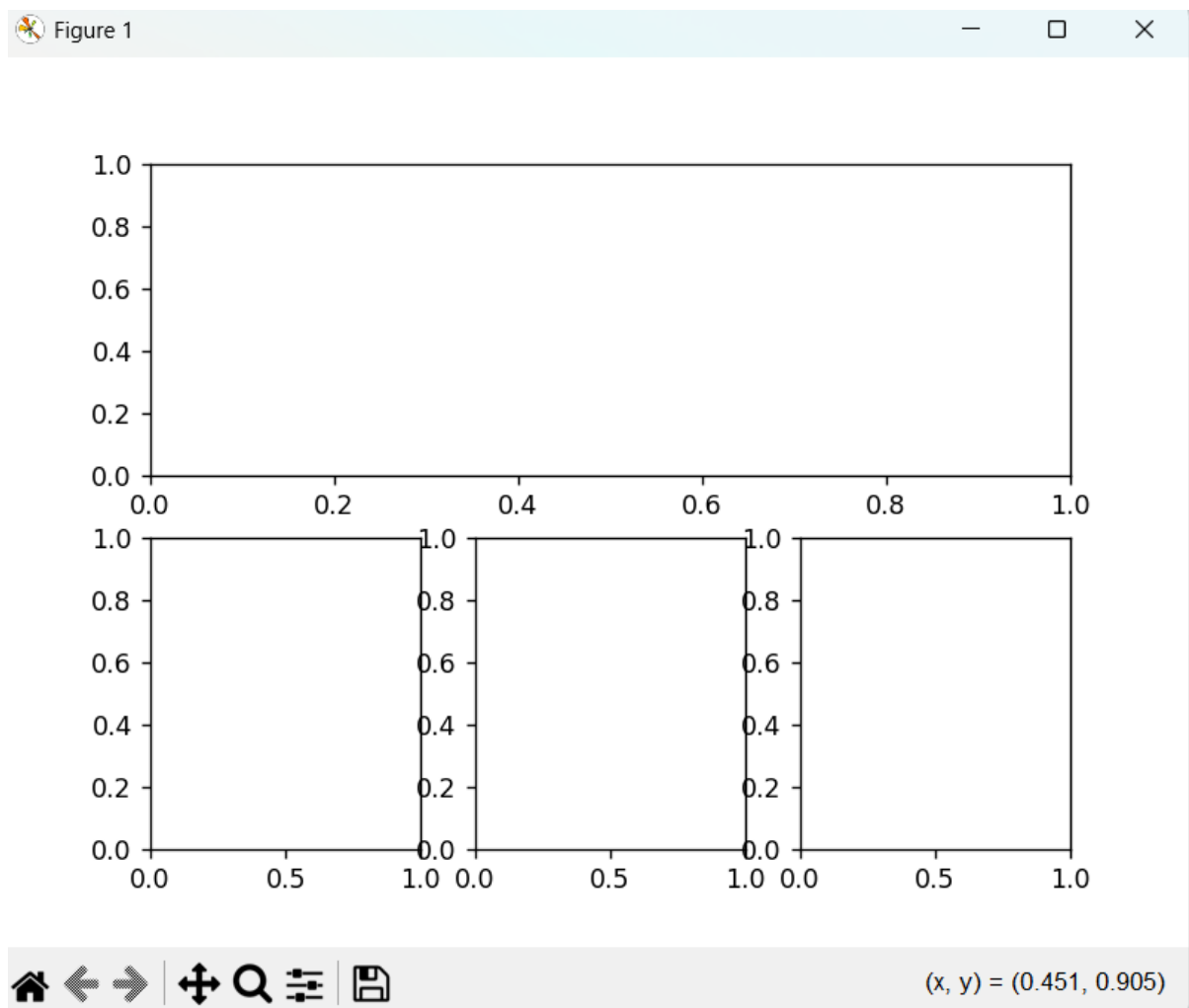
1. Import the matplotlib.pyplot module.
2. Create a figure object.
3. Add multiple subplot areas using add_subplot().
4. Display the subplots in different positions.
5. Show the figure using plt.show().

Program

Python Code

```
import matplotlib.pyplot as plt
fig = plt.figure()
fig.add_subplot(2, 1, 1)
fig.add_subplot(2, 3, 4)
fig.add_subplot(2, 3, 5)
fig.add_subplot(2, 3, 6)
plt.show()
```

Output



Result

Thus, the Python program to create multiple plots in a single figure was successfully executed and the multiple plots were displayed.

EXP 27

Aim

To write a Python program to display a bar chart of the popularity of programming languages.

Algorithm

1. Import the matplotlib.pyplot module.
2. Create a list of programming languages.
3. Create a list of their popularity values.
4. Draw a bar chart using plt.bar().
5. Add labels to the X-axis and Y-axis.
6. Add a title to the chart.
7. Display the chart using plt.show().

Program

Python Code

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8.0, 7.7, 6.7]
```

```
plt.bar(languages, popularity)
```

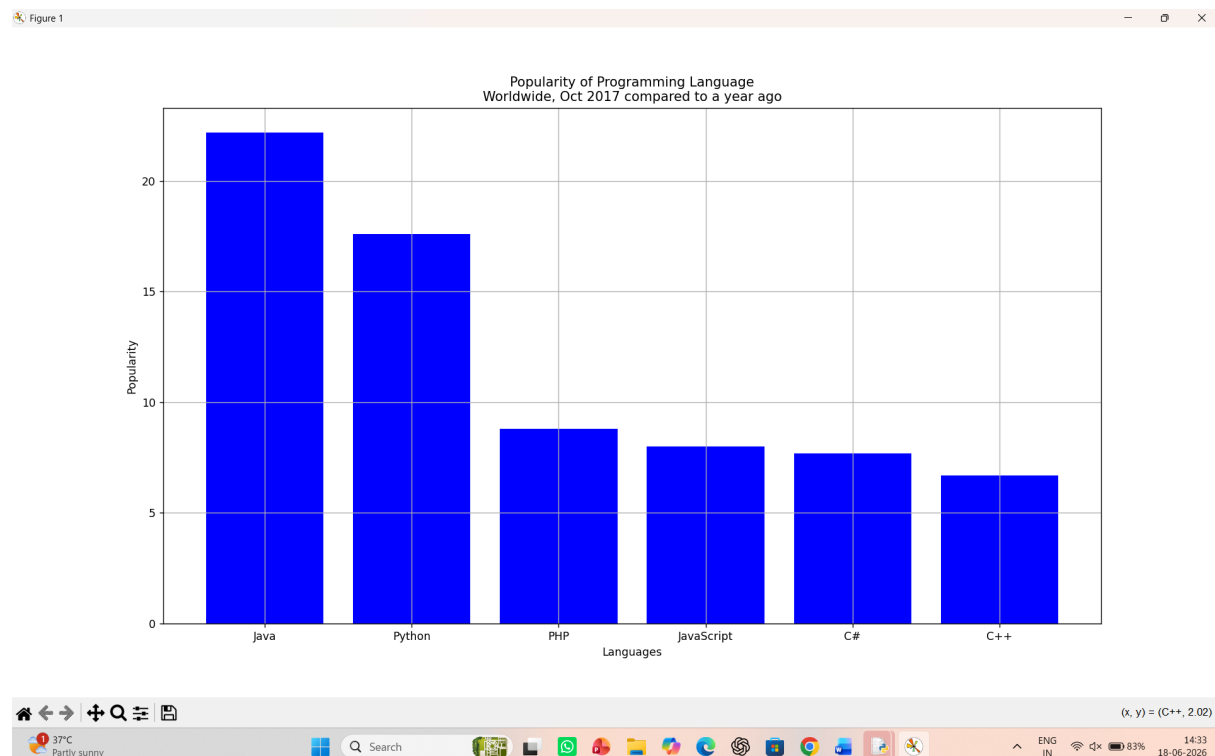
```
plt.xlabel('Languages')
```

```
plt.ylabel('Popularity')
```

```
plt.title('Popularity of Programming Language\nWorldwide, Oct 2017 compared to a year ago')
```

```
plt.show()
```

Output



Result

Thus, the Python program to display a bar chart of the popularity of programming languages was successfully executed and the bar chart was displayed.

EXP 28

Aim

To write a Python program to display a horizontal bar chart of the popularity of programming languages.

Algorithm

1. Import the matplotlib.pyplot module.
2. Create a list of programming languages.
3. Create a list of popularity values.
4. Draw a horizontal bar chart using plt.barh().
5. Add labels to the X-axis and Y-axis.
6. Add a title to the chart.
7. Display grid lines.
8. Show the chart using plt.show().

Program

Python Code

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'Python', 'PHP', 'JS', 'C#', 'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8.0, 7.7, 6.7]
```

```
plt.barh(languages, popularity, color='green')
```

```
plt.xlabel('Popularity')
```

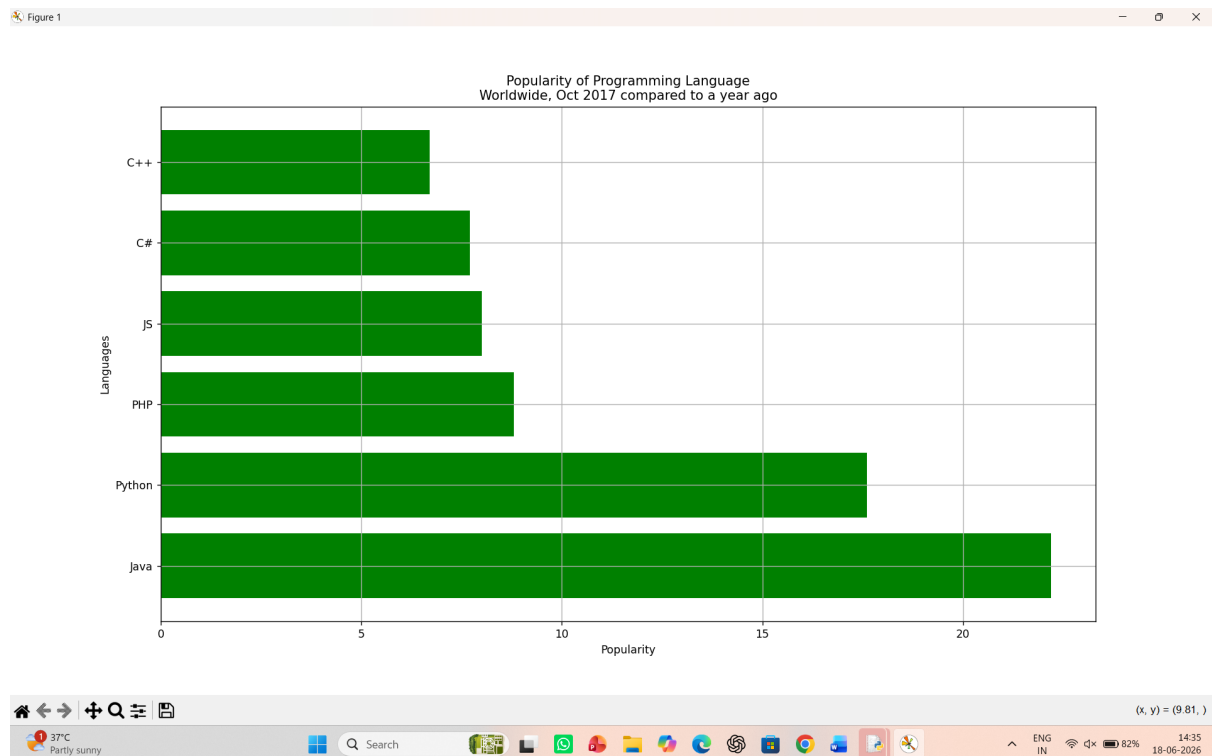
```
plt.ylabel('Languages')
```

```
plt.title('Popularity of Programming Language\nWorldwide, Oct 2017 compared to a year ago')
```

```
plt.grid(True)
```

```
plt.show()
```

Output



Result

Thus, the Python program to display a horizontal bar chart of the popularity of programming languages was successfully executed and the chart was displayed.

EXP 29

Aim

To write a Python program to display a bar chart of the popularity of programming languages using different colors for each bar.

Algorithm

1. Import the matplotlib.pyplot module.
2. Create a list of programming languages.
3. Create a list of popularity values.
4. Specify different colors for each bar.
5. Draw a bar chart using plt.bar().
6. Add labels to the X-axis and Y-axis.
7. Add a title to the chart.
8. Display grid lines.
9. Show the chart using plt.show().

Program

Python Code

```
import matplotlib.pyplot as plt
```

```
languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
```

```
popularity = [22.2, 17.6, 8.8, 8.0, 7.7, 6.7]
```

```
colors = ['red', 'black', 'green', 'blue', 'yellow', 'cyan']
```

```
plt.bar(languages, popularity, color=colors)
```

```
plt.xlabel('Languages')
```

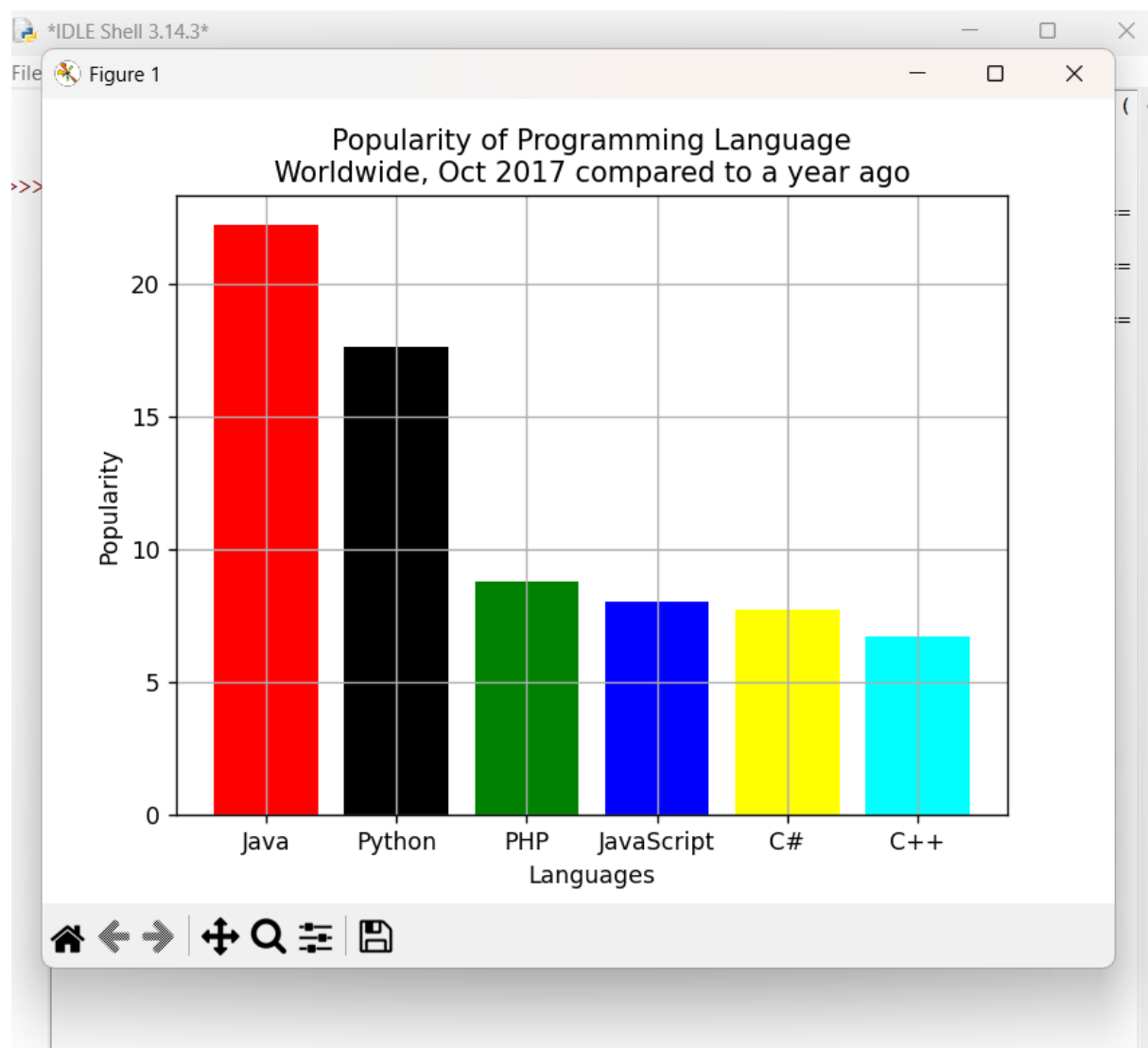
```
plt.ylabel('Popularity')
```

```
plt.title('Popularity of Programming Language\nWorldwide, Oct 2017 compared to a year ago')
```

```
plt.grid(True)
```

```
plt.show()
```

Output



Result

Thus, the Python program to display a bar chart of the popularity of programming languages using different colors for each bar was successfully executed and the chart was displayed.

EXP 30

Aim

To write a Python program to create a bar plot of scores by group and gender using multiple X values on the same chart for men and women.

Algorithm

1. Import the required modules: matplotlib.pyplot and numpy.
2. Create the data for men and women scores.
3. Define the group labels.
4. Set the positions of the bars on the X-axis.
5. Plot the bars for men and women with different colors.
6. Add labels to the X-axis and Y-axis.
7. Add a title and legend.
8. Display the chart using plt.show().

Program

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

men_means = (22, 30, 33, 30, 26)
women_means = (25, 32, 30, 35, 29)

groups = ['G1', 'G2', 'G3', 'G4', 'G5']

x = np.arange(len(groups))
width = 0.35

plt.bar(x, men_means, width, color='green', label='Men')
```

```
plt.bar(x + width, women_means, width, color='red', label='Women')
```

```
plt.xlabel('Person')
```

```
plt.ylabel('Scores')
```

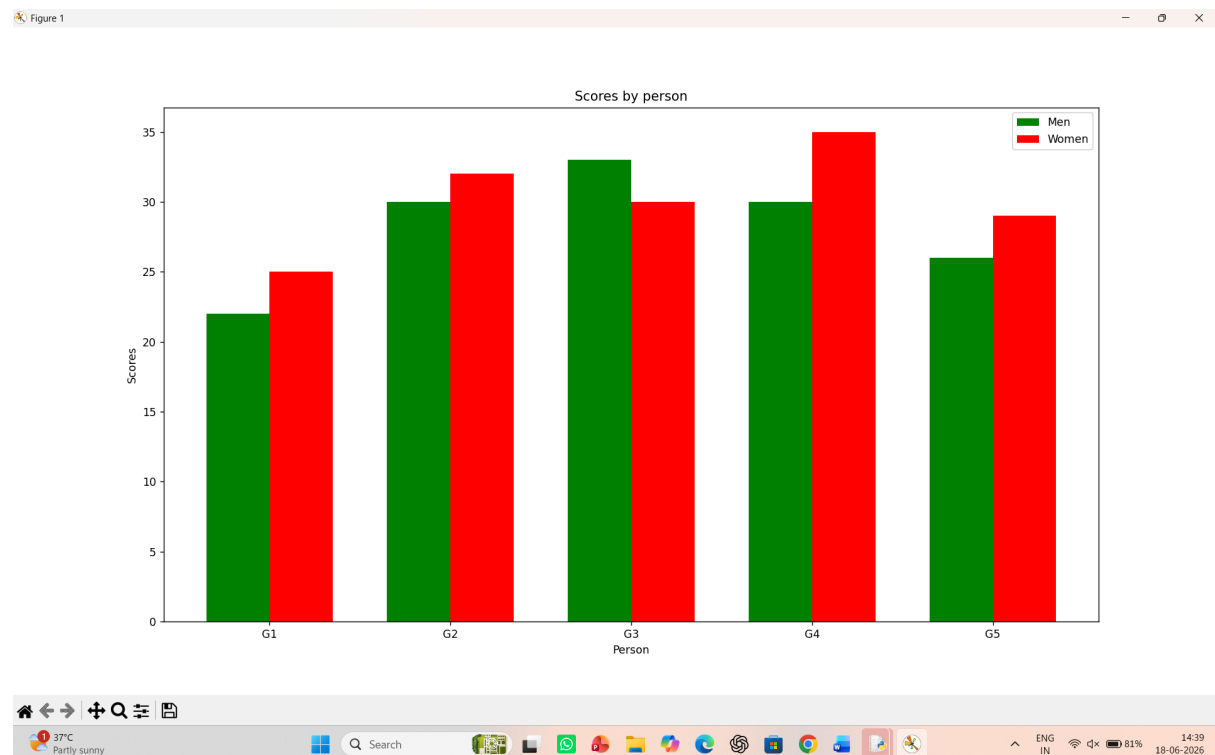
```
plt.title('Scores by person')
```

```
plt.xticks(x + width/2, groups)
```

```
plt.legend()
```

```
plt.show()
```

Output



Result

Thus, the Python program to create a bar plot of scores by group and gender using multiple X values on the same chart for men and women was successfully executed and the chart was displayed.

